OBJECT ORIENTED SOFTWARE ENGINEERING

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Pattern Categories

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Patterns Categories

we group patterns into **three** **categories**:

- **Architectural patterns**
- **Design patterns**
- **Idioms**

- Each category consists of patterns having a similar range of scale or abstraction.

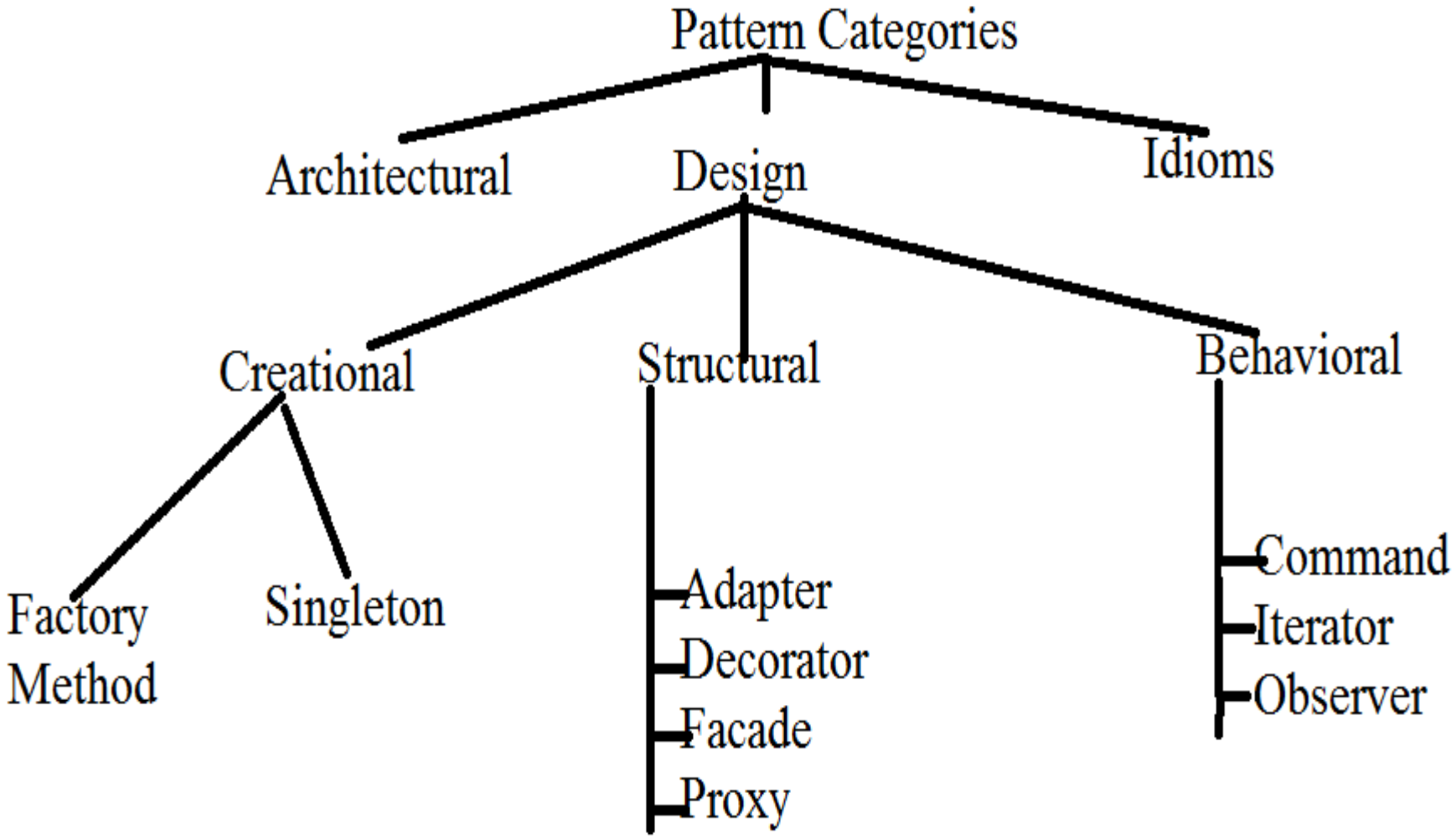
Design Pattern - Types of Design Pattern

As per the design pattern reference book **Design Patterns - Elements of Reusable Object-Oriented Software , there are 23 design patterns.**

These 23 patterns can be classified in **three categories:**

1. **Creational**
2. **Structural**
3. **Behavioral patterns**

Design Pattern



1. Architectural Patterns

- Architectural patterns are used to describe **viable software architectures** that are built according to some overall structuring principle.

Definition:

- An architectural pattern expresses **a fundamental structural organization schema for software systems.**

- It provides a set of **predefined subsystems, specifies their responsibilities, and includes rules and guidelines** for organizing the relationships between them.

- Example: **Model-view-controller pattern (MVC)**

- Design patterns are used to describe **subsystems of a software architecture as well as the relationships between them** (which usually consists of several smaller architectural units)

Definition:

- A design pattern provides a scheme for refining the subsystems or components of a software system, or the relationships between them.

Patterns

- It describes a **commonly-recurring structure of communicating components** that solves a general design problem within a particular Context.

- They are **medium-scale patterns**. They are smaller in scale than architectural patterns, but tend to be independent of a particular programming language or programming paradigm.

- Example: **Publisher-Subscriber pattern**

3. Idioms

- Idioms deals with the **implementation of particular design issues**.

Definition:

- An idiom is a **low-level pattern** **specific to a programming language**.
- An idiom describes how to implement particular aspects of components or the relationships between them using the features of the given language.
- Idioms represent the **lowest- level patterns**. They address aspects of both design and implementation.
- Eg: **Counted body pattern**

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Pattern Description

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Pattern Description

1. **Name** : The **name** and a short summary of the pattern.

2. **Also known as**: **Other names** for the pattern, if any are known.

3. **Example** : A real world example demonstrating the **existence of the problem** and the need for the pattern.

4. **Context** :The **situations** in which the patterns may apply.

5. **Problem** : The problem the pattern addresses, including a discussion of its associated forces.

6. Solution : The fundamental solution principle underlying the pattern.

7. Structure : A detailed specification of the structural aspects of the pattern, including **CRC** – cards for each participating component and an **OMT class diagram**.

8. Dynamics : Typical scenarios describing the run time behavior of the pattern.

9. Examples resolved: Discussion for any important aspects for resolving the example that are not yet covered in the solution , structure, dynamics and implementation sections.

Pattern Description…

10. Variants: A brief description of variants or **specialization** of a pattern

11. Known uses: Examples of the use of the pattern, taken from existing systems.

12. Consequences: The benefits the pattern provides, and any potential liabilities.

13. See Also: References to patterns that solve similar problems, and the patterns that help us refine the pattern we are describing

THANK YOU

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